OmniFlow

Agentic Alexa+ Life & Commerce Copilot • Technical Architecture & Submission Whitepaper

Amazon Developer Hackathon 2026

Executive Summary: OmniFlow is an autonomous, multi-modal Alexa+ Agent Skill and self-hosted Model Context Protocol (MCP) server compliant with MCP specification version **2025-11-25 over Streamable HTTP**, integrated with **Amazon Bedrock (Claude 3.5 Sonnet)**. It transitions digital assistants from single-turn reactive bots into proactive, goal-driven orchestrators capable of resolving high-level complex tasks across services with deterministic safety guardrails.

1. Primary Track & Challenge Alignments

Category	Specification / Target	Implementation Details
Primary Track	Alexa+ Agent Skill & MCP Server	Fully compliant with MCP 2025-11-25 over Streamable HTTP (SSE); multi-modal visual cards on Echo Show & Web Simulator.
Mini-Challenge 1	AWS Builder Mini Challenge	Integrated Amazon Bedrock Runtime SDK (`@aws-sdk/client-bedrock-runtime`) with Claude 3.5 Sonnet Converse API function calling.
Mini-Challenge 2	Open Source Mini Challenge	100% open-source under MIT License at `https://github.com/akshatanand-cell/omniflow-alexa-mcp`.
Bonus Points	Product Feedback & Friction Log (+10%)	Comprehensive friction log covering Streamable HTTP SSE header negotiation and Bedrock schema validation in `docs/FRICTION_LOG.md`.

2. System Architecture & Model Context Protocol Tools

OmniFlow exposes four atomic tools conforming to JSON-Schema parameters exported directly via standard JSON-RPC 2.0 endpoints:

- **`plan_itinerary`**: Formulates comprehensive multi-day trip schedules, activities, outdoor locations, and weather forecasts.
- **`search_amazon_products`**: Live product search querying Prime-eligible outdoor gear with verified reviews, star ratings, and prices.
- **`optimize_budget_cart`**: Reconciles cart items against user-defined budget ceilings, calculating bundle discounts (-$12.50) and estimated tax.
- **`confirm_order_execution`**: Enforces human-in-the-loop authorization tokens, generating secured one-touch approval cards before checkout.

3. Amazon Bedrock Autonomous Reasoning Engine

Autonomous Goal Decomposition

When presented with complex goals like *"Plan a 3-day camping trip to Olympic National Park under $300 and buy needed gear"*, Bedrock deconstructs high-level requirements into parallel tool actions without human micro-management.

Deterministic Safety Guardrails

Financial and commerce commitments cannot execute autonomously without explicit human-in-the-loop confirmation. The agent constructs staged carts and renders a high-visibility approval card.

4. Production Deployment & Verification

- **Production Live URL**: https://omniflow-alexa-mcp.vercel.app/
- **MCP Protocol Discovery Endpoint**: `https://omniflow-alexa-mcp.vercel.app/mcp/info`
- **Code Repository**: https://github.com/akshatanand-cell/omniflow-alexa-mcp
- **Demo Video (YouTube)**: https://youtu.be/frLnS64f22E